

ESP32 ADAPTER SAFETY PROTOTYPE

Complete Hardware Connections — Current Working Build

This document is for the current ESP32 prototype. It includes the DS18B20 replacement for the LM35, the three tested fault-simulation potentiometers, LCD, LEDs, buzzer, relay, ACS712, voltage sensor, and buttons.

SAFETY: Use only an isolated low-voltage DC test source. Never connect 230 V AC mains to the breadboard, ESP32, sensors, LCD, or GPIO pins. Use the relay only with a low-voltage test load.

Verified/current behavior

Part	Current result
POT1/POT2/POT3	Working; values change with rotation
16×2 I2C LCD	Working; address 0x27
LEDs	Green GPIO2, Yellow GPIO4, Red GPIO15
Buzzer	GPIO25; HIGH = ON, LOW = OFF
Relay	GPIO26; tested HIGH = ON
DS18B20	GPIO19 with 4.7 kΩ DATA pull-up
ACS712	Wired; calibration still required
0–25 V sensor	Wired; calibration still required

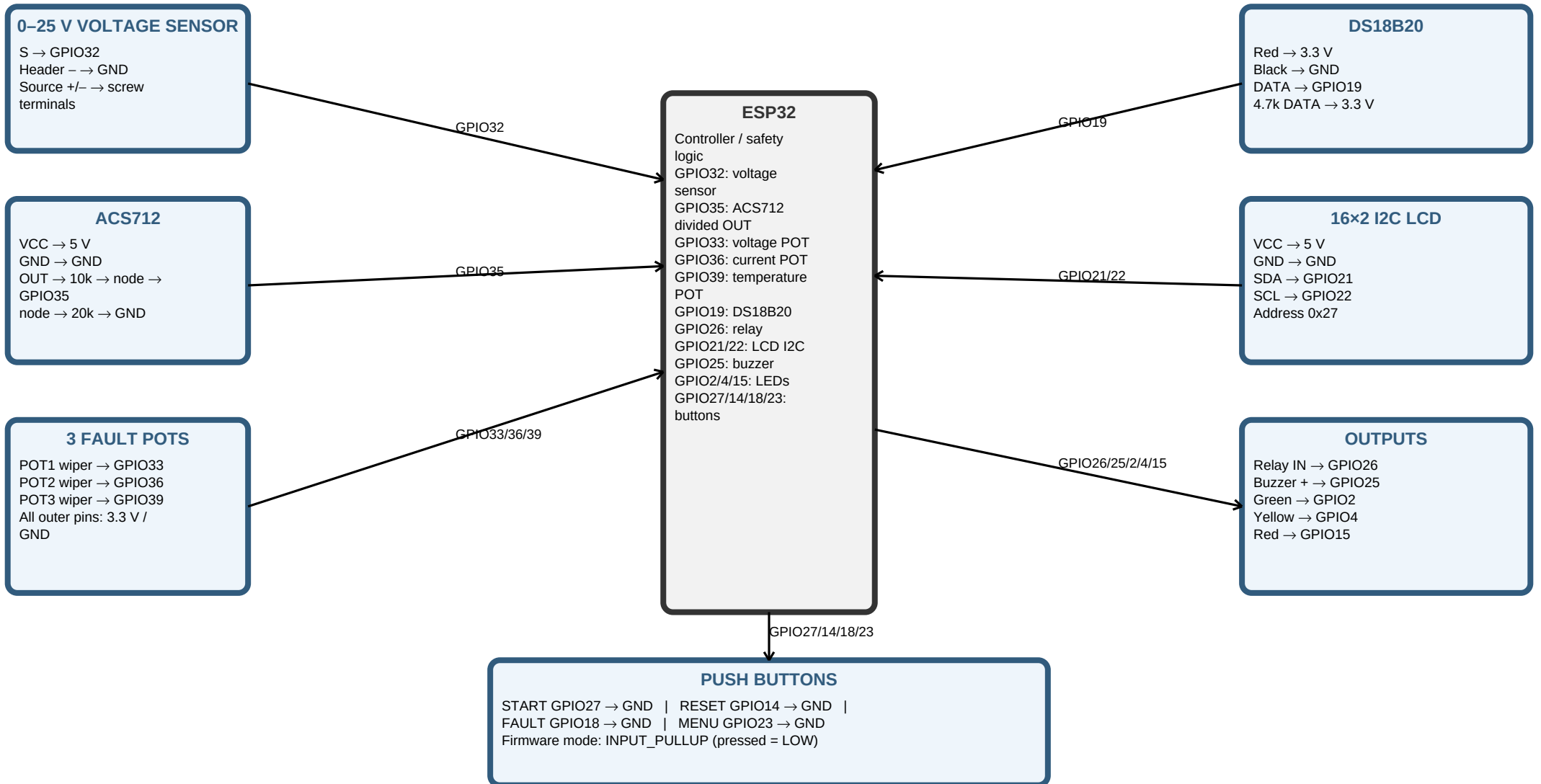
ESP32 ADAPTER SAFETY PROTOTYPE — CIRCUIT DIAGRAM

Current tested build • low-voltage prototype only • GPIO26 relay HIGH = ON • GPIO25 buzzer HIGH = ON

3.3 V RAIL

5 V RAIL

COMMON GND



SAFETY: No 230 V AC mains. Relay contacts are for an isolated low-voltage test load only.

DETAILED WIRING DIAGRAM — POWER, SENSORS & OUTPUTS

Use this page together with the pin-by-pin connection tables in the main PDF.

3.3 V

5 V

GND

POTENTIOMETERS

Each pot:
Outer pin A → 3.3 V
Middle/wiper → ADC GPIO
Outer pin B → GND
POT1 → GPIO33
POT2 → GPIO36 / VP
POT3 → GPIO39 / VN
Never use 5 V on pot ends.

DS18B20

RED → 3.3 V
BLACK → GND
DATA → GPIO19
4.7 kΩ: DATA → 3.3 V

ACS712 → ESP32 ADC

ACS VCC → 5 V
ACS GND → GND
ACS OUT → 10 kΩ → NODE
NODE → GPIO35
NODE → 20 kΩ → GND
Low-voltage current path:
Source + → IP+ → IP- → load +
Load - → source -

0–25 V SENSOR

Source + → screw +
Source - → screw -
S → GPIO32
Header - → GND
Header + → leave open

16×2 I2C LCD

VCC → 5 V
GND → GND
SDA → GPIO21
SCL → GPIO22
Address: 0x27
Direct SDA/SCL was
tested successfully.

STATUS LEDs

GREEN: GPIO2 → 220Ω → LED → GND
YELLOW: GPIO4 → 220Ω → LED → GND
RED: GPIO15 → 220Ω → LED → GND
Use one resistor per LED.

BUZZER

+ → GPIO25
- → GND
Tested:
HIGH = ON
LOW = OFF

RELAY MODULE

VCC → 5 V
GND → GND
IN → GPIO26
Tested:
HIGH = ON
LOW = OFF
COM/NO/NC → low-voltage
test load only

BUTTONS

START → GPIO27 → GND
RESET → GPIO14 → GND
FAULT → GPIO18 → GND
MENU → GPIO23 → GND
INPUT_PULLUP

ESP32 CONNECTION SUMMARY

GPIO32 Voltage | GPIO35 ACS712 | GPIO33/36/39 Pots
GPIO19 DS18B20 | GPIO21/22 LCD | GPIO25 Buzzer
GPIO26 Relay | GPIO2/4/15 LEDs
GPIO27/14/18/23 Buttons

LOW-VOLTAGE ONLY: Do not connect any 230 V AC source to the prototype.

1. MASTER ESP32 PIN MAP

Use this as the main reference while wiring.

Function	GPIO	Connection / note
Voltage sensor S	32	S → GPIO32
ACS712 divided OUT	35	OUT → 10 kΩ → node; node → GPIO35 and 20 kΩ → GND
POT1 voltage fault	33	Wiper → GPIO33; ends → 3.3 V/GND
POT2 current fault	36 / VP	Wiper → GPIO36; ends → 3.3 V/GND
POT3 temperature fault	39 / VN	Wiper → GPIO39; ends → 3.3 V/GND
DS18B20 DATA	19	DATA → GPIO19; 4.7 kΩ DATA → 3.3 V
Relay IN	26	Tested: HIGH = ON
LCD SDA	21	LCD SDA → GPIO21
LCD SCL	22	LCD SCL → GPIO22
Buzzer +	25	+ → GPIO25; – → GND; HIGH = ON
Green LED	2	GPIO2 → 220 Ω → anode; cathode → GND
Yellow LED	4	GPIO4 → 220 Ω → anode; cathode → GND
Red LED	15	GPIO15 → 220 Ω → anode; cathode → GND
START	27	Other terminal → GND; INPUT_PULLUP
RESET	14	Other terminal → GND; INPUT_PULLUP
FAULT TEST	18	Current build; other terminal → GND
MENU	23	Other terminal → GND; INPUT_PULLUP

2. POWER RAILS & COMMON GROUND

Keep 3.3 V and 5 V rails separate. The ESP32 and low-voltage modules share a common GND.

Rail	Connect to	Important
3.3 V	All three pots; DS18B20; 4.7 k Ω pull-up	Do not power the pots from 5 V.
5 V	ACS712 VCC; relay VCC; LCD VCC	Use only the appropriate supply for the actual module.
GND	All module grounds; LED cathodes; buzzer –; button grounds	Common low-voltage ground.

Recommended setup: connect ESP32 3.3 V to one breadboard rail, ESP32 GND to the ground rail, and keep the 5 V rail separate. Connect every listed GND to the common ground rail.

Do not connect mains voltage to any breadboard rail.

3. THREE FAULT-SIMULATION POTENTIOMETERS

Each pot simulates one fault quantity. The current simplified firmware maps its ADC reading to 0–100%.

Pot	Purpose	Outer 1	Wiper	Outer 2
POT1	Voltage fault	3.3 V	GPIO33	GND
POT2	Current fault	3.3 V	GPIO36	GND
POT3	Temperature fault	3.3 V	GPIO39	GND

The two outer pins may be reversed if you want the knob direction reversed. The middle pin must go to the listed GPIO.

4. DS18B20 TEMPERATURE SENSOR

This is the current replacement for the burned LM35.

DS18B20 wire	Connect to
Red	ESP32 3.3 V
Black	ESP32 GND
Yellow / White DATA	ESP32 GPIO19
4.7 k Ω resistor	Between GPIO19/DATA and 3.3 V

Resistor placement: one resistor leg goes to the DATA/GPIO19 node and the other goes to 3.3 V. Never connect DATA to 5 V.

5. ACS712 CURRENT SENSOR

ACS712 point	Connection
VCC	5 V
GND	Common GND
OUT	OUT $\rightarrow$ 10 k Ω $\rightarrow$ measurement node
Measurement node	Node $\rightarrow$ GPIO35 AND node $\rightarrow$ 20 k Ω $\rightarrow$ GND
Current path	Low-voltage source + $\rightarrow$ IP+ $\rightarrow$ IP- $\rightarrow$ load +; load - $\rightarrow$ source -

Current measurement needs calibration against a known load before treating the displayed value as accurate.

6. 0–25 V VOLTAGE SENSOR

Sensor connection	Connect to
Measured source +	Module screw terminal + / VCC
Measured source –	Module screw terminal – / GND
S	ESP32 GPIO32
Header –	ESP32 GND
Header +	Leave unconnected on the common module used here

Use only low-voltage DC. Do not add another external 10 k Ω /10 k Ω divider on top of this common 0–25 V module. Calibrate against a multimeter.

7. 16×2 I2C LCD

The LCD scanner found address **0x27**. The direct wiring below is the arrangement already tested successfully.

LCD pin	ESP32 connection
VCC	5 V
GND	GND
SDA	GPIO21
SCL	GPIO22
I2C address	0x27

The workshop guide recommends level shifting for the LCD. The direct SDA/SCL connection above is the one already tested successfully in this build.

8. STATUS LEDs

LED	GPIO	Exact wiring
Green	2	GPIO2 → 220 Ω → LED long leg/anode; short leg/cathode → GND
Yellow	4	GPIO4 → 220 Ω → LED long leg/anode; short leg/cathode → GND
Red	15	GPIO15 → 220 Ω → LED long leg/anode; short leg/cathode → GND

Use a separate 220 Ω resistor for each LED.

9. BUZZER

Buzzer terminal	Connection
+	GPIO25
–	GND

The standalone test confirmed: **GPIO25 HIGH = buzzer ON**; LOW = OFF.

10. RELAY MODULE

Relay pin	Connection	Current tested behavior
VCC	5 V	Module power
GND	Common GND	Module ground
IN	GPIO26	HIGH = ON ; LOW = OFF, based on your test
COM/NO/NC	Low-voltage test load only	Confirm labels on the physical module

Note: Some relay modules are active-LOW, but your connected module was tested as HIGH = ON. Use the tested result for the current code.

11. LOW-VOLTAGE RELAY LOAD WIRING

For a normally disconnected low-voltage test load, use:

Source + → Relay COM → Relay NO → Test-load +

Source – → Test-load –

Use COM + NO if you want the load disconnected when the relay is not energized. Confirm COM/NO/NC markings on the actual relay before connecting the load.

Never use this breadboard prototype to switch 230 V AC.

12. FOUR PUSH BUTTONS

Button	GPIO	Other terminal	Mode
START	27	GND	INPUT_PULLUP
RESET	14	GND	INPUT_PULLUP
FAULT TEST	18	GND	INPUT_PULLUP — current build
MENU	23	GND	INPUT_PULLUP

For a 4-pin tactile switch, place it across the breadboard center gap. One electrically connected side goes to the GPIO and the opposite side goes to GND.

13. CURRENT PROTOTYPE LOGIC

The simplified firmware uses the three pots as simulated voltage/current/temperature fault levels. The highest percentage controls the overall state.

Highest pot	Status	Green	Yellow	Red	Buzzer	Relay
0–49 %	SAFE	ON	OFF	OFF	OFF	ON
50–79 %	WARNING	OFF	ON	OFF	OFF	ON
80–100 %	FAULT	OFF	OFF	ON	ON	OFF

LCD format: V : xx% C : xx% on line 1 and T : xx% SAFE/WARNING/FAULT on line 2.

14. STEP-BY-STEP TEST PROCEDURE

1. Power the ESP32 from USB only and verify the board is stable.
2. Confirm the LCD displays correctly at address 0x27.
3. Turn each pot and verify its percentage changes.
4. Below 50% → green. 50–79% → yellow. 80%+ → red.
5. Buzzer should be OFF in SAFE/WARNING and ON in FAULT.
6. Relay should be ON in SAFE/WARNING and OFF in FAULT in the current firmware.
7. Test ACS712 and voltage sensor separately before relying on their numerical values.
8. If using relay contacts, use only an isolated low-voltage DC test load.

15. FINAL WIRING CHECKLIST

Item	Verify
Common ground	ESP32 GND connected to module grounds, LED cathodes, buzzer –, button grounds
3.3 V rail	All three pots and DS18B20 use 3.3 V
POT1/POT2/POT3	GPIO33 / GPIO36 / GPIO39 respectively
DS18B20	Red 3.3 V; black GND; DATA GPIO19; 4.7 k Ω DATA→3.3 V
ACS712	5 V/GND; 10 k Ω series to node; node GPIO35; 20 k Ω node→GND
Voltage sensor	S→GPIO32; header →→GND; header + unconnected
LCD	5 V/GND; SDA21; SCL22; address 0x27
LEDs	GPIO2/4/15 with separate 220 Ω resistors
Buzzer	+→GPIO25; →→GND; HIGH=ON
Relay	VCC5V; GND; IN GPIO26; HIGH=ON as tested
Buttons	START27, RESET14, FAULT18, MENU23; other terminal to GND
Mains	NO 230 V AC CONNECTIONS

16. QUICK ONE-LINE REFERENCE

POT1: 3.3V – wiper GPIO33 – GND

POT2: 3.3V – wiper GPIO36 – GND

POT3: 3.3V – wiper GPIO39 – GND

DS18B20: red 3.3V – black GND – DATA GPIO19 + 4.7k to 3.3V

ACS712: 5V/GND; OUT–10k–node–GPIO35; node–20k–GND

Voltage: source +/- to screw terminals; S GPIO32; header – GND

LCD: 5V/GND; SDA21; SCL22; 0x27

LEDs: GPIO2/4/15 → 220 Ω → anode; cathode → GND

Buzzer: + GPIO25; – GND; HIGH=ON

Relay: VCC 5V; GND; IN GPIO26; HIGH=ON

Buttons: START27, RESET14, FAULT18, MENU23 → other side GND

This document is based on the project wiring guide plus the current tested build changes: DS18B20 on GPIO19, FAULT button on GPIO18, direct LCD wiring, buzzer HIGH=ON, and relay HIGH=ON as tested.